

Jun (Keith) Yang

Last updated on Sep. 14, 2026

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Education

- 2024 – 2029 **Ph.D.** (Quantitative Biosciences), **Georgia Institute of Technology**
(expected) Advisor: Dr. Hannah Choi
- 2020 – 2024 **B.Sc. & B.Eng.**, **Tsinghua University**
Major: Mathematics and Physics + Electrical Engineering and Automation

Research Interests

- Neural dynamics
- Statistical field theory for neural networks
- Predictive coding

Publications

Preprints

- [1] **Yang, J.** & Choi, H. (2026). Nonlinear dynamics of random neural networks with second-order synaptic motifs. *arXiv*, <https://doi.org/10.48550/arXiv.2609.14251>

Journal Articles

- [1] **Yang, J.**, Zhang, H. & Lim, S. (2024). Sensory-memory interactions via modular structure explain errors in visual working memory. *eLife* **13**, RP95160. <https://doi.org/10.7554/eLife.95160.4>

Conference Abstracts

- [2] **Yang, J.** & Choi, H. (2026). Nonlinear dynamics of random neural networks with second-order connectivity motifs. *COSYNE*, Program No. 3-028.
- [1] **Yang, J.**, Zhang, H. & Lim, S. (2023). Cardinal repulsion in working memory requires sensory-memory network interactions. *SfN Neuroscience*, Program No. 169.03.

Permanent Preprints

- [1] **Yang, J.** (2025). Theories on random recurrent neural networks: a brief review. *OSF Preprints*, https://doi.org/10.31219/osf.io/ztfn7_v4

Summer Schools and Workshops

- 2025 **Modeling Software Workshop**, Allen Institute
A workshop on BMTK and VND.

- 2024 **CNeuro 2024**, Tsinghua University
A one-week computational neuroscience summer school.
- 2023 **The 12th Computational Neuroscience Winter School**, Online
A winter school organized by Shanghai Jiao Tong University

Scholarships & Awards

- 2021 – 2023 **Scholarship of Scientific or Technological Innovation Excellence**
Tsinghua University
- 2020 – 2022 **Scholarship of Academic Excellence**
Tsinghua University

Teaching

Teaching assistantship (at Georgia Tech)

Term	Course	Duty
2025 Fall	MATH 4221 Stochastic Processes I	Grader
2025 Fall	MATH 4581 Math Methods in Engr	Grader
2025 Summer	MATH 1553 Intro to Linear Algebra	Taught studio sessions
2025 Spring	MATH 1553 Intro to Linear Algebra	Taught studio sessions (i.e., recitations)
2024 Fall	MATH 1554 Linear Algebra	Grader

Technical skills

	Skill	Level	Detail
Programming/ Typesetting	C		First programming language learned.
	Python		For data analysis and machine learning (BMTK, AllenSDK, PyNest, NumPy, scikit-learn, CVXPY, PyTorch, Matplotlib, IDTxl, etc.).
	MATLAB		Main tool for simulation and data analysis. MatCont, MatPower, MINT.
	Julia		Main tool for high-performance scientific computing.
	Wolfram Mathematica		Beginner.
	LaTeX & Typst		Typesetting academic papers.
Softwares	Microsoft Office Suite		PowerPoint, Word, Excel, OneNote, etc.

Languages	Skill	Level	Detail
	Adobe Photoshop & Illustrator		Making figures for academic papers.
	Git		Code version management.
	English		Fluent in academic speech and writing.
	Chinese		Native language.
basic knowledge			extensive knowledge
intermediate knowledge			expert knowledge